RedFlag Lab 1 Outline

Team Bronze

CS 411W

4 April 2026

Table of Contents

1 Introduction 3

2 RedFlag Product Description 3

3 Glossary 4

4 References 4

Listing of Figures

Figure 1: Misinformation Growth Graph 3

Figure 2: RedFlag UI Flag Example 3

1 Introduction

- Misinformation spreads rapidly across online platforms, including social media and news websites, making it difficult for users to distinguish between reliable and misleading content

- Research from a 2018 MIT study shows that false information spreads significantly faster than truthful information

- Users currently lack real-time tools to verify content while browsing, forcing them to rely on self judgment

- Current solutions typically require manual fact-checking or require users to redirect to external websites, introducing friction and discourages consistent use

- Gap: There's no passive real-time system that detects and flags misinformation directly within the browsing experience

- [Figure 1: Graph showing misinformation growth]

- Introduce RedFlag: A background system that automatically identifies potential misinformation, flags it for users, and provides multiple perspectives to support informed decision making.

2 RedFlag Product Description

RedFlag is a fact–checking and bias indicator utility that works across multiple platforms. It runs in the background and scans text to find any inaccuracies and provide sources for the user to look into. This information and sources provided give the user a way to look further into the misinformation. There is also the BiasStar which will add a star to any politically divisive content and offers further viewpoints from sources across the political spectrum.

[Figure 2: RedFlag High Level Concept Diagram]

2.1 Key Features

- Real-Time Integrated Detection
    - Feature: Scans the user's screen to find any misinformation.
    - Significance: Works automatically in the background without requiring user input
    - Addresses: Tackles the inconvenience of the user having to leave their current browsing experience by bringing sources directly to them
- Highlight Warnings
    - Feature: Highlights any questionable information on the users' browsing experience with no user input

- Significance: Does not intrude on the users' experience and presents the highlight subtly
- Addresses: Attempts to intervene before the user shares misinformation, while not entirely interrupting the experience

- Balanced Perspectives Engine (BiasStar)
  - Feature: Identifies political content and provides multiple viewpoint sources across the political spectrum
  - Significance: Doesn't label ideologies explicitly but offers an evaluation from every perspective
  - Addresses: Tackles how social media may reinforce bias and echo chambers
- On Demand Explanations
  - Feature: Detailed overviews are provided on the flagged information and explained to the user
  - Significance: Allows the user to make a more information decision without dictating how the user should think on the topic
  - Addresses: Provides better credibility towards content and allows the user to think further on the information

2.2 Major Components

[Figure 3: Major Functional Component Diagram (MFCD)]

- Presentation Layer (Frontend)
  - Components: The RedFlag browser extension
  - Function: Displays real time alerts and a view panel for users
  - Tech Stack: Built with Javascript for browser extensions
- Application Layer (Backend)
  - Components: Content detection, Misinformation Analysis, Fact Verification, Balanced Perspectives Engine, and Alert & Recommendation Generator
  - Tech Stack: Python (Flash/Django) for the information datasets
- Data Layer (Database)
  - Components: PostgreSQL database
  - Stored Data: Source Credibility Database (Includes reliability scores, and political alignments), User Interaction Data, and Content Intelligence Data (Known false claims, topic classifications)
- External Interfaces / APIs
  - Fact checking API: Compares questionable information against the database
  - News API: Takes public news at input and categorizes by topic

- LLM API: Parses database information to summarize its findings and increase fact-checking speed

3 Identification of Case Study

For whom is the product being developed? Why? Who else might use this in the future?

- WHO: 16 year-old teenager, Daiskue, spends large amounts of his free time scrolling through his phone on Instagram

- PROBLEM: He recognizes some false information on his Instagram feed, but doesn't want to spend the time researching every post he comes across.

- FEATURES: Redflag misinformation indicators

- SCENARIO: Daiskue enables Redflag on his browser, and indicators pop up on questionable text. He receives multiple sources of information on questionable topics, and is able to go to those sources immediately.

- FUTURE: Word of mouth, discussion of Redflag with his friends

4 Glossary

- Misinformation: False or misleading information

- API: Application Programming Interface

- NLP: Natural Language Processing

- OCR: Optical Character Recognition

4 References

- APA format